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Complex Engineering Problems (CEP)

Attributes

Addressing the Complex Engineering Problems (P) in the Project

The project required knowledge of Database Management Systems (DBMS), SQL, Java programming, ER diagrams, and database relationships. We also learned how to connect the frontend with the database and used Git for version control and localhost for testing during development.

P1: Depth of knowledge required

P2: Range of conflicting requirements

The project required dividing the work among team members while ensuring that all parts of the system worked together correctly. Coordination between database development, frontend design, and backend integration was necessary to complete the project successfully.

P3: Depth of analysis required

We analyzed the project requirements before development and used knowledge from previous courses to design the database, connect different system components, and solve problems that arose during implementation and testing.

P4: Familiarity of issues

The project involved applying concepts such as Java programming, Object-Oriented Programming (OOP), problem-solving, data structures, SQL queries, and database design in a real application, which helped us become more familiar with these topics. The project depends on the interaction between the frontend,

P7: Interdependence backend, database, Git, and localhost. Proper integration of these components was necessary for the system to function correctly.

Complex Engineering Activities (CEA)

Attributes

Addressing the Complex Engineering Activities (A) in the Project

The project involved both human and technical resources. Team members were responsible for different parts of the project, while

A1: Range of

resources tools such as MySQL, Git, Visual Studio Code, html and localhost were used for development, testing, and project management.

Attributes	Addressing the Complex Engineering Activities (A) in the Project
A2: Level of interactions	Team members worked together by sharing responsibilities and coordinating their tasks. Regular communication and collaboration helped integrate the database, frontend, and backend into a complete system.
A3: Innovation	The project provides a platform where students can exchange skills without spending money. Instead of paying for courses, users can teach a skill they know and learn a different skill from others, encouraging collaboration and knowledge sharing.
A4: Consequences to society / Environment	The platform encourages learning through peer-to-peer skill sharing, making education more accessible and affordable. It also promotes cooperation among students and strengthens the learning community.
A5: Familiarity	The system is designed so that users with basic computer knowledge can easily use its features. Students can register, browse skills, and exchange knowledge through a simple and user-friendly interface.